

6. Connection interface

A Connection is a session between a Java application and a database. It helps to establish a connection with the database. The Connection interface is a factory of Statement, PreparedStatement, and DatabaseMetaData, i.e., an object of Connection can be used to get the object of Statement and DatabaseMetaData. The Connection interface provide many methods for transaction management like commit(), rollback(), setAutoCommit(), setTransactionIsolation(), etc.

Note: By default, connection commits the changes after executing queries.

6.1 Commonly used methods of Connection interface

1) public Statement createStatement():

creates a statement object that can be used to execute SQL queries.

2) public Statement createStatement(int resultSetType,int resultSetConcurrency):

Creates a Statement object that will generate ResultSet objects with the given type and concurrency.

3) public void setAutoCommit(boolean status):

is used to set the commit status. By default, it is true.

4) public void commit():

saves the changes made since the previous commit/rollback is permanent.

5) public void rollback():

Drops all changes made since the previous commit/rollback.

6) public void close():

closes the connection and Releases a JDBC resources immediately.

6.2 Connection Interface Fields

There are some common Connection interface constant fields that are present in the Connect interface. These fields specify the isolation level of a transaction.

TRANSACTION_NONE: No transaction is supported, and it is indicated by this constant.

TRANSACTION_READ_COMMITTED: It is a constant which shows that the dirty reads are not allowed. However, phantom reads and non-repeatable reads can occur.

TRANSACTION_READ_UNCOMMITTED: It is a constant which shows that dirty reads, non-repeatable reads, and phantom reads can occur.

TRANSACTION_REPEATABLE_READ: It is a constant which shows that the non-repeatable reads and dirty reads are not allowed. However, phantom reads can occur.

TRANSACTION_SERIALIZABLE: It is a constant which shows that the non-repeatable reads, dirty reads as well as the phantom reads are not allowed.